Phase 1: Classical Machine Learning Pipeline for Facial Skin Condition Classification

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***Abstract*—This paper presents the Phase 1 baseline for six-class facial skin-condition classification using classical machine learning exclusively. Each image is resized to 224×224 pixels and encoded as a 154-dimensional handcrafted descriptor built from HSV colour histograms, Local Binary Patterns (LBP), and Gray-Level Co-occurrence Matrix (GLCM) statistics. This descriptor is evaluated with an RBF-kernel Support Vector Machine (SVM) and a Random Forest on the same stratified split of $8{,}823$ processed images. The Random Forest yields the best result: 0.809 accuracy and 0.806 weighted F1. The analysis shows that colour features are most informative for rosacea and dark spots, while texture features are most informative for eczema and wrinkles; acne remains the most difficult class because localised lesions are diluted by global handcrafted representations. This explainable baseline provides a reference point for comparison against the deep learning and hybrid fusion models in subsequent phases.**

***Index Terms*—computer vision, skin condition classification, handcrafted features, local binary patterns, gray-level co-occurrence matrix, support vector machine, random forest**

I. Introduction

Facial skin-condition classification is challenging because several classes share overlapping colour and texture cues. The six target classes in this project are *normal*, *acne*, *wrinkles*, *eczema*, *rosacea*, and *dark spots*. Even when the class labels are clinically intuitive, the visual differences can be subtle under changes in lighting, pose, and image quality.

This report covers Phase 1 only and intentionally operates within classical machine learning. The objective is to build an interpretable baseline before introducing deep representations in Phase 2 and hybrid fusion in Phase 3, a phased strategy supported by the broader dermatology literature's strong emphasis on learned image models [1]. Each image is resized, normalised, and converted into a compact feature vector that combines colour and texture information. Two standard classifiers are then evaluated on the same stratified split: an RBF-kernel SVM [2] and a Random Forest [3].

The main finding is that handcrafted features provide a competitive six-class baseline. The Random Forest reaches 0.809 accuracy and 0.806 weighted F1 on the held-out test set, while the failure analysis reveals where global descriptors break down and motivates the transition to spatially richer representations in later phases.

II. Related Work

The literature underlying this baseline spans roughly five decades and divides naturally into three groups: (A) handcrafted texture and colour descriptors, (B) classical classifiers, and (C) domain-level evidence from dermatology computer vision. Reviewing them in this order is not merely organisational convenience: it mirrors the historical progression from engineered features to learned representations, and it makes explicit the gap that Phase 1 is designed to occupy.

A. Handcrafted Texture and Colour Descriptors

Haralick, Shanmugam, and Dinstein [4] introduced the Gray-Level Co-occurrence Matrix in 1973, defining fourteen second-order statistics — including contrast, homogeneity, energy, and correlation — that capture spatial co-occurrence structure beyond the reach of single-pixel histograms. For over two decades, GLCM was the dominant texture representation in medical image analysis. Its key limitation, however, is one the authors themselves do not address: the original formulation is defined at a single fixed distance d and a single angle θ. Multi-distance, multi-angle GLCM is a widely adopted extension [4] that only emerged in subsequent applied work. This matters for skin analysis: wrinkle lines and eczema scaling vary in spatial frequency, so a single-distance descriptor systematically under-represents texture at coarser scales. The present work uses $d \in \{1, 2\}$ and $\theta \in \{0°, 45°, 90°, 135°\}$ precisely to compensate for this gap in the foundational paper.

Ojala, Pietikäinen, and Mäenpää [5] formalised the rotation-invariant, multiresolution Local Binary Pattern operator nearly three decades later. Where Haralick's GLCM captures coarser second-order co-occurrence structure, LBP encodes the sign pattern of intensity differences in a local neighbourhood, making it sensitive to micro-texture at the pixel level. Crucially, LBP is rotation-invariant by construction: the uniform-pattern histogram is invariant to rigid image rotation, a property GLCM does not share without explicit angular averaging. For clinical skin photography, which is captured at arbitrary orientations, this distinction is non-trivial. LBP is therefore not simply an alternative to GLCM — the two descriptors are complementary: LBP targets micro-texture and is rotation-invariant; GLCM targets macro-texture at multiple scales and is orientation-sensitive. The Phase 1 pipeline concatenates both, capturing texture information that neither descriptor could provide alone.

Wei, Gan, and Ji [6] provided the closest direct precedent for combining colour and GLCM texture features in a skin-condition classification pipeline. Using an SVM classifier on a three-class dataset of herpes, dermatitis, and psoriasis, they showed that colour-texture fusion outperformed either modality in isolation on all three classes. Two limitations

bound the applicability of this result to the present work. First, the three target diseases do not overlap with the six classes studied here; the finding that colour-texture fusion is beneficial for pigmentation-driven conditions *transfers by analogy* to rosacea and dark spots, but is not a direct empirical claim about those conditions. Second, the paper does not report per-class precision and recall, making it impossible to determine whether the fusion benefit is uniformly distributed across classes or concentrated in conditions where one modality is individually weak. This gap motivates the per-class analysis in Section VI.

B. Classical Classifiers

Cortes and Vapnik [2] introduced the Support Vector Machine as a maximum-margin binary classifier, and demonstrated that non-linear boundaries could be learned via the kernel trick without explicitly constructing the high-dimensional feature mapping. The RBF kernel $K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2)$ is the natural choice for a 154-dimensional mixed feature space where the relevant decision boundary is unlikely to be linear. A critical limitation, however, is that the original theoretical analysis addresses binary classification exclusively; the six-class extension used here employs a one-vs-one strategy, which carries no comparable margin-maximisation guarantee. This approximation may explain the SVM's relatively weaker performance on acne and rosacea, where the multi-class boundary is likely irregular.

Breiman [3] showed that aggregating many decorrelated decision trees — each trained on a random feature subset at every split — yields ensemble classifiers that are robust on heterogeneous, mixed-scale predictor spaces and moderately class-imbalanced datasets. Both properties are directly relevant to Phase 1: the 154-dimensional descriptor combines colour histograms (continuous, globally distributed) with LBP and GLCM statistics (discrete, locally structured), a heterogeneity that single-boundary classifiers like SVM handle with a single global kernel. Random Forest, by contrast, constructs localised tree boundaries that can model interactions *within* and *across* the colour and texture sub-vectors independently.

Comparing these two classifiers directly on the same hand-crafted feature vector is itself a methodological contribution of Phase 1: it isolates the effect of classifier architecture on a fixed representation, providing a cleaner ablation signal than any comparison that jointly varies both the features and the classifier.

C. Dermatology Computer Vision: The State of the Art and the Gap

Kassem et al. [1] conducted a systematic review of 102 papers on skin lesion classification and diagnosis, published between 2015 and 2021. Their findings define the current state of the art: deep convolutional networks — primarily ResNet, VGG, and EfficientNet variants — achieve classification accuracies of 0.89–0.97 on the ISIC 2018 benchmark, an approximately 8–15 percentage-point advantage over the strongest classical baselines cited in that review.

The Kassem review, however, identifies three structural weaknesses in the existing deep learning literature that the present project directly addresses. First, the reviewed datasets are dominated by a small number of large benchmarks and show limited skin-tone diversity, raising unresolved questions about cross-demographic generalisation. Second, the reviewed systems are predominantly end-to-end learned, providing no interpretable intermediate representation — a limitation that becomes acute in clinical settings where practitioners must justify automated predictions. Third, and most relevant to this project, *no widely adopted explainable classical baseline exists for multi-class facial skin condition classification* that spans the six conditions studied here; the review's classical baselines are restricted to binary or three-class settings.

This is the precise gap Phase 1 fills. Deep learning pipelines outperform handcrafted baselines on established benchmarks — that is not in dispute. The question this project poses is different: *how much of that performance advantage persists when deep and classical models are evaluated on the same six-class split, with the same training data, under the same experimental protocol?* Without a well-constructed classical baseline, that question cannot be answered. Phase 1 constructs that baseline; Phases 2 and 3 answer the question.

Taken together, these three groups of papers lead to a single design conclusion: the right Phase 1 pipeline combines HSV colour histograms (motivated by Wei et al. [6]'s colour-texture fusion finding), LBP micro-texture and GLCM macro-texture (motivated by their complementarity established by Ojala et al. [5] and Haralick et al. [4] respectively), and a Random Forest classifier (motivated by Breiman's [3] robustness on heterogeneous feature spaces), with an SVM for comparison (motivated by Cortes and Vapnik [2]) that isolates the classifier effect on a fixed representation. This design is not the product of arbitrary choice; it is the consequence of reading the literature's contributions and limitations together.

III. Dataset Description and Exploratory Analysis

The data loader accepts folder-based class labels and normalises common aliases such as `dark_spots` and `class0_normal`. After filtering unreadable files, the processed cache contains 8,823 images across six classes. A stratified 80/20 split yields 7,058 training samples and 1,765 test samples.

TABLE I
Processed Dataset Size by Class

Class	Total	Train	Test
Acne	1,185	948	237
Dark spots	1,057	846	211
Eczema	3,133	2,506	627
Normal	1,240	992	248
Rosacea	1,108	886	222
Wrinkles	1,100	880	220
Total	8,823	7,058	1,765

Table I shows that the class distribution is imbalanced: eczema accounts for 35% of all images while dark spots accounts for only 12%. This 2.96× ratio is not extreme enough to block learning, but is strong enough to distort accuracy as a standalone metric. Weighted precision, recall, and F1 are therefore reported throughout.

The EDA stage guided preprocessing rather than serving merely as a display of sample images. Because handcrafted features are sensitive to spatial geometry, every image is resized to 224×224 pixels and normalised to $[0, 1]$. A secondary observation is that the dataset is condition-centric rather than selfie-centric: images typically show isolated lesion regions against relatively clean backgrounds, which benefits global feature extraction but limits direct transfer to natural phone photos.

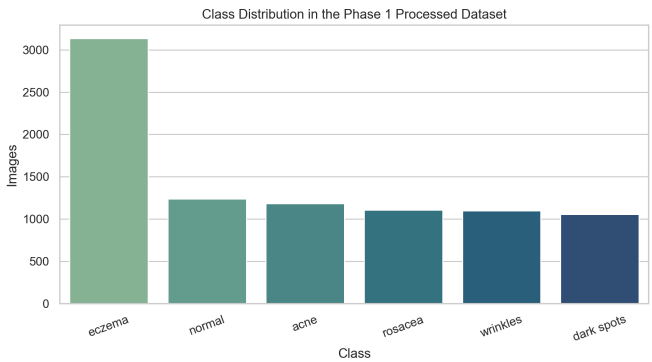

Fig. 1. Class distribution of the 8,823-image processed dataset. Eczema dominates at 3,133 images; dark spots is the smallest class at 1,057. Weighted evaluation metrics are used throughout to account for this imbalance.

IV. Methodology

Each image is loaded with OpenCV, resized to 224×224 pixels, and normalised to $[0, 1]$. Corrupted images are skipped rather than allowed to terminate the pipeline. The final descriptor concatenates three complementary feature blocks into a 154-dimensional vector:

$$\underbrace{3 \times 32}_{\text{HSV}} + \underbrace{(24 + 2)}_{\text{LBP}} + \underbrace{(4 \times 2 \times 4)}_{\text{GLCM}} = 154.$$

A. HSV Colour Histograms

The HSV histogram captures chromatic structure. For channel c, the normalised histogram is

$$h_c(b) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\big[I_c(i) \in \mathcal{B}_b\big], \tag{1}$$

where N is the number of pixels and $\mathcal{B}_b$ is the b-th bin. HSV is preferred over RGB because hue and saturation are more closely aligned with the redness and hyperpigmentation cues relevant to rosacea and dark spots, respectively. Each channel uses 32 bins, yielding $3 \times 32 = 96$ values.

B. Local Binary Patterns

LBP captures micro-texture via the operator defined by Ojala et al. [5]:

$$\mathrm{LBP}_{P,R} = \sum_{p=0}^{P-1} s(g_p - g_c)\, 2^p, \tag{2}$$

where g_c is the central pixel intensity, g_p are the P neighbours on a circle of radius R, and $s(\cdot)$ is the sign function. The parameters $P = 24$, $R = 3$ are chosen to capture fine lines and surface roughness relevant to eczema and wrinkles. Uniform LBP produces $P + 2 = 26$ histogram bins.

C. GLCM Statistics

GLCM [4] complements LBP by summarising coarser spatial co-occurrence structure. Four statistics — contrast, homogeneity, energy, and correlation — are extracted at distances $d \in \{1, 2\}$ and angles $\theta \in \{0°, 45°, 90°, 135°\}$, giving $4 \times 2 \times 4 = 32$ values. Using two distances captures texture at different spatial scales, which the original Haralick formulation at a fixed single distance does not address.

D. Classifiers

Two classical classifiers are compared. The SVM uses the RBF kernel [2]:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\!\big(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\big), \tag{3}$$

applied after z-score standardisation of the 154-dimensional descriptor. The Random Forest [3] uses 300 trees with balanced class weights. The pair is intentionally minimal and interpretable: SVM provides a strong non-linear margin boundary; Random Forest is robust on heterogeneous, mixed-scale feature spaces such as the combination of colour histograms and texture statistics used here.

V. Experimental Setup

The pipeline is implemented in Python 3 with OpenCV, NumPy, pandas, scikit-learn, matplotlib, and seaborn. Extracted features are cached to `data/processed/features_dataset.csv` so that later experiments can reuse the same Phase 1 representation without re-running the extractor.

TABLE II
Hyperparameters for Both Classifiers

Parameter	SVM	Random Forest
Kernel / criterion	RBF	Gini impurity
C / n_estimators	1.0 (default)	300
γ	`scale`	—
Class weight	balanced	balanced
Probability output	enabled	—
Standardisation	z-score	none
`test_size = 0.20`, `random_state = 42`		

Both models are trained on the same stratified split (`random_state = 42`), ensuring that all class proportions

are preserved across train and test partitions. Because eczema dominates the class distribution, weighted precision, recall, and F1 are emphasised alongside accuracy in all comparisons. The evaluation stage exports confusion matrices, per-class performance plots, and serialised model files (`.pkl`) so that the same evidence is reusable in Phase 2 and Phase 3 ablation tables.

VI. Results and Discussion

Table III shows that the Random Forest is the stronger overall baseline, achieving 0.809 accuracy and 0.806 weighted F1 compared with 0.785 accuracy and 0.788 weighted F1 for the SVM.

TABLE III
PHASE 1 BASELINE COMPARISON ON THE SHARED TEST SPLIT
($n = 1{,}765$)

Model	Acc.	Prec.	Recall	F1
Random Forest	0.809	0.811	0.809	0.806
SVM (RBF)	0.785	0.800	0.785	0.788

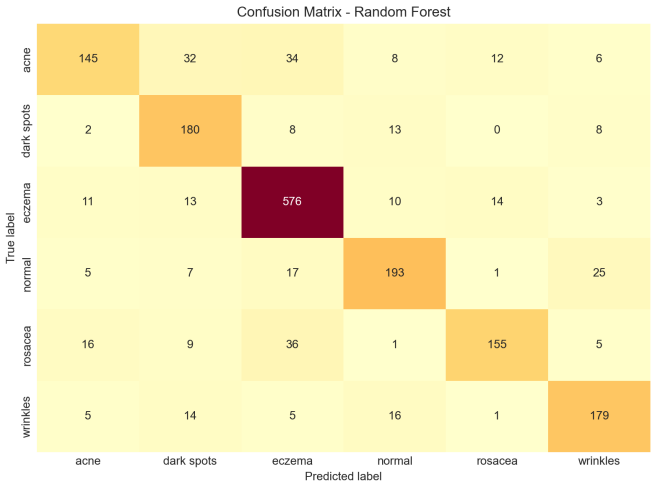

Fig. 2. Random Forest confusion matrix on the Phase 1 test split. The diagonal shows correct predictions per class. Eczema accounts for the largest absolute count (576 correct out of 627; recall = 0.919) owing to its class size. Acne shows the highest off-diagonal mass: 32 true-acne samples are misclassified as dark spots and 34 as eczema.

A. Per-Class Analysis

The performance gap between models is driven primarily by eczema and acne. For eczema — the largest class, representing 35% of the test set — the Random Forest achieves recall = 0.919 versus 0.887 for SVM, a 0.032 gap that disproportionately lifts the Random Forest's weighted score. Eczema benefits from LBP micro-texture cues (surface roughness and scaling) and moderate colour homogeneity, both of which align well with the global descriptor design.

For acne, both models struggle: the Random Forest records recall = 0.612 and F1 = 0.689, the weakest per-class result. Acne lesions are spatially localised — small papules and pustules that occupy only a fraction of the 224×224 image — so their discriminative signal is diluted by the surrounding unaffected skin in the global descriptor. The SVM achieves marginally higher acne recall (0.631) because the margin-based objective is more sensitive to the minority region of the SVM decision boundary, but this advantage does not compensate for its weaker eczema performance.

Dark spots are handled well by both classifiers (RF F1 = 0.884) because reddish-brown pigmentation produces distinctive peaks in the Hue and Saturation channels of the HSV histogram (Eq. 1), and this signal is largely global across the image patch. Wrinkles similarly benefit from the LBP histogram (Eq. 2) capturing fine-line micro-texture.

The Random Forest outperforms the SVM on the mixed 154-dimensional feature space because its localised decision rules model interactions between the colour and texture sub-vectors more effectively than a single global kernel boundary [3]. This makes it the natural Phase 1 model to carry forward into the ablation study.

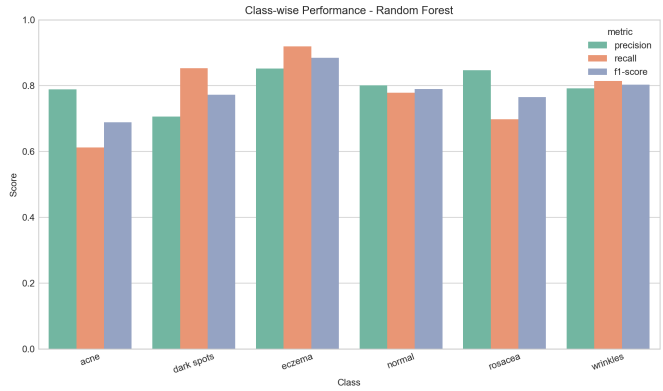

Fig. 3. Random Forest per-class precision, recall, and F1 on the test split. Eczema, wrinkles, and dark spots show the strongest performance. Acne (F1 = 0.689) is the weakest major class, a consequence of the global descriptor's inability to resolve spatially localised lesions.

VII. Failure Analysis and Limitations

A. Global Descriptor and Localised Lesions

The principal limitation of this baseline is that it summarises the entire resized image with a single global feature vector. This approach is effective for conditions that produce broad, spatially distributed signatures — eczema, dark spots, rosacea — but weakens substantially for acne, where small lesions ($\lesssim 10\%$ of image area) are averaged together with unaffected skin. The result is a recall of 0.612 and F1 of 0.689 for acne under the best model, the lowest of any class.

B. Selfie-Style Generalisation

The same structural limitation makes selfie-style inference harder than condition-centric dataset evaluation. A selfie introduces background content, hair, specular highlights, and variable pose that can dominate the global colour and texture

statistics, shifting the feature vector away from its training distribution. The model should therefore be understood as a condition-centric Phase 1 baseline rather than a deployment-ready system.

C. Other Limitations

Three additional limitations bound this phase. First, no face-region segmentation is applied during training; the full 224×224 crop is used, including hairline and neck regions that carry no diagnostic information. Second, hyperparameter tuning is limited to a single default configuration; a grid search over SVM C, γ, and RF `n_estimators` would likely improve both models. Third, no comparison against deep or hybrid models is yet available; that gap defines the motivation for Phases 2 and 3.

VIII. Conclusion and Future Work

This Phase 1 baseline demonstrates that a 154-dimensional handcrafted descriptor combining HSV (Eq. 1), LBP (Eq. 2), and GLCM statistics [4] can support competitive six-class skin condition classification while remaining fully interpretable. Among the two models evaluated, the Random Forest achieves 0.809 accuracy and 0.806 weighted F1 and is the recommended baseline to carry into the ablation study.

The experiments also define the precise boundary of this approach: global descriptors are effective for texture- and colour-distributed conditions but fail to capture spatially localised pathology, as evidenced by acne recall of 0.612. This boundary is the justification for Phase 2 (CNN transfer learning with MobileNetV2/EfficientNet) and Phase 3 (attention-weighted hybrid fusion), both of which are designed to overcome the locality limitation while preserving the interpretability contributions established here. The feature cache produced in Phase 1 is directly reusable as one input stream to the Phase 3 fusion model.

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